

Theoretical and Computational Mapping of Musical Emotion: A Multimodal Framework for Affective Trajectory Analysis

The systematic mapping of musical listening data to emotional and thematic patterns over time requires a rigorous, empirically grounded taxonomy of musical affect. Because music conveys and elicits complex emotional states, a computational model designed for automated analysis must reconcile the differences between general psychological models of emotion and domain-specific musical frameworks. This report evaluates the leading taxonomies in music psychology, establishes an optimal hybrid framework, presents a standardized eight-category emotion taxonomy, provides diagnostic criteria for resolving ambiguous emotional boundaries, and outlines the primary theoretical limitations of the proposed architecture.

Comparative Analysis of Musical Emotion Taxonomies

A comparative analysis of the primary emotion frameworks is essential to identify the optimal model for computational music information retrieval. These frameworks represent diverse theoretical lineages, ranging from general-purpose dimensional models to domain-specific, music-induced scales.

Russell's Circumplex Model of Affect

The Circumplex Model represents emotions as continuous coordinates within a low-dimensional space defined by two orthogonal axes: Valence (hedonic tone, ranging from pleasure to displeasure) and Arousal (physiological activation, ranging from sleepiness to excitement). The continuous nature of this model makes it mathematically convenient for algorithmic systems, enabling real-time trajectory tracking and distance-based similarity calculations. However, the model struggles to differentiate between distinct emotional states that occupy similar quadrants. For example, anger and fear reside in the high-arousal, negative-valence quadrant, while peacefulness and sadness reside in the low-arousal half of the space. This poor resolution limits the model's ability to capture the aesthetic nuances of musical expression.

Juslin & Laukka's Basic Emotions Framework

This framework focuses on discrete, universal, and biologically innate categories—typically happiness, sadness, anger, fear, and tenderness. The core theoretical strength of this model lies in its evolutionary grounding. Acoustic analyses show that musicians convey emotions using the same patterns of acoustic cues—such as pitch height, tempo, intensity, and duration—that humans use in vocal prosody to communicate affect. While basic emotions are recognized with high cross-cultural accuracy and are easily computed from audio signals, they do not account for the complex aesthetic experiences of music, such as nostalgia, wonder, or transcendence,

which are central to user engagement.

Zentner's Geneva Emotional Music Scale and GEMIAC

The Geneva Emotional Music Scale (GEMS-45, GEMS-25, GEMS-9) is a domain-specific model designed to measure emotions induced by music. It identifies nine primary dimensions: wonder, transcendence, tenderness, nostalgia, peacefulness, power, joyful activation, tension, and sadness. While GEMS captures complex aesthetic emotions more accurately than general-purpose scales, it has significant computational drawbacks: the dimensions are statistically redundant, and several (such as transcendence and wonder) are too abstract for automated classification.

To address these limitations, the Geneva Music-Induced Affect Checklist (GEMIAC) was developed. GEMIAC utilizes a fuzzy-set approach, representing each emotional class with a pair of semantically overlapping terms to capture subtle variations. GEMIAC developers empirically tested the inclusion of basic emotions like fear and surprise. Fear terms did not cluster independently, with one term merging into tension and others failing to cluster, while surprise terms clustered under the wonder branch. To correct the positive-valence bias of GEMS, GEMIAC added boredom as a validated, independent low-activation negative category and refined sadness and tension.

Eerola & Vuoskoski's Dimensional-Discrete Synthesis

This framework systematically integrates discrete emotion categories with dimensional models of affect, confirming a high correspondence between discrete ratings and the valence and arousal axes. Their research demonstrates that while a two-dimensional space captures the majority of variance in musical affect, discrete categories struggle to represent emotionally ambiguous or transitional stimuli. The synthesis establishes that the dimensional model is highly effective for characterizing continuous emotional transitions, whereas discrete categories are best suited for representing prototypical, high-intensity musical moments.

Cowen & Keltner's 13-Dimensional Semantic Space

Using large-scale statistical tools, Cowen and Keltner identified 13 distinct dimensions of subjective experience associated with music across Western and Chinese cultures: amusement, joy, eroticism, beauty, relaxation, sadness, dreaminess, triumph, anxiety, scariness, annoyance, defiance, and feeling pumped up. This model shows that the subjective experience of music is high-dimensional and organized along continuous gradients rather than isolated categories. While highly rigorous and descriptive of modern popular music, managing 13 continuous dimensions introduces significant computational complexity for real-time trajectory tracking.

Eerola et al.'s EPISODE Model

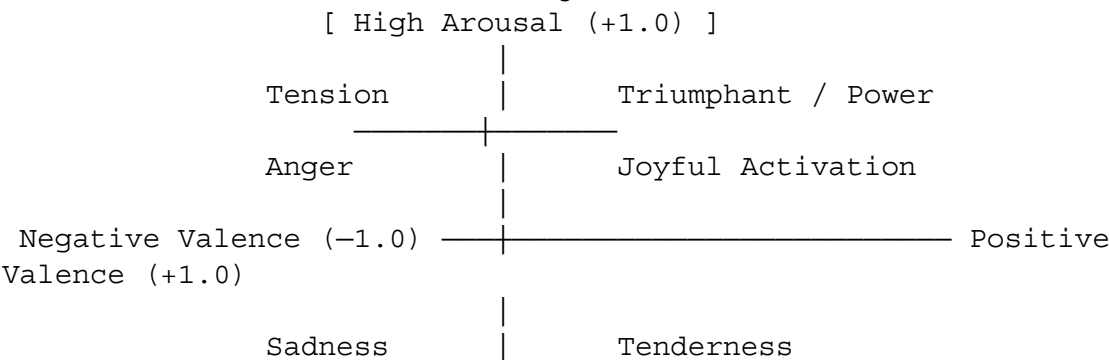
The EPISODE model is a situation-based framework that places the listener's context and goals at the center of the experience. It defines five functionally unique episodes of emotional music engagement: Enjoyment-Distracton-Relaxation, Connection-Belonging, Focus-Motivation, Personal Emotional Processing, and Aesthetic-Interest-Awe. Rather than classifying the acoustic properties of a track, the EPISODE model categorizes the functional relationship between the listener and the music, providing a framework for mapping thematic user behavior

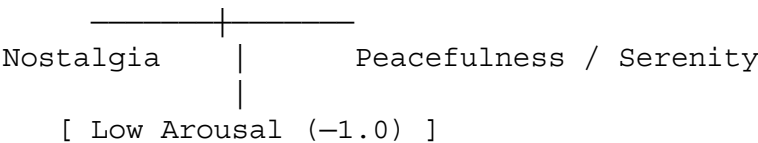
over time.

Framework	Core Dimensions / Categories	Empirical Strength	Technical Limitation
Russell's Circumplex	Valence, Arousal	Universal cognitive baseline; mathematically simple.	Poor discriminability for distinct emotions in the same quadrant.
Juslin & Laukka	Joy, Sadness, Anger, Fear, Tenderness	Standardized acoustic cues; evolutionary vocal prosody validation.	Fails to capture complex, non-utilitarian aesthetic emotions.
Zentner's GEMS	9 Dimensions; 3 Meta-factors	Domain-specific; highly accurate for classical art music.	Computational redundancy; positive bias; abstract dimensions.
GEMIAC	GEMS + Boredom; refined Tension/Sadness	Fuzzy-set approach; validated negative categories; empirical pruning.	High coordinate complexity; semantic term dependency.
Eerola & Vuoskoski	Synthesized Discrete & Dimensional	Bridges the discrete-dimensional divide; robust acoustic mapping.	Primarily focused on audio signals rather than lyrical semantics.
Cowen & Keltner	13 Interconnected Dimensions	Large-scale cross-cultural validation; highly descriptive of modern music.	High dimensionality increases computational overhead in tracking.
EPISODE Model	5 Functional Episodes	Situation-centric; highly effective for tracking user intent.	Cannot be extracted purely from acoustic or lyrical features alone.

The Optimal Hybrid Framework: Technical Rationale

To map listening data to emotional trajectories, the recommended framework is a hybrid architecture: a continuous two-dimensional Valence-Arousal (VA) coordinate system populated by eight discrete, music-specific categories. This design combines the mathematical tractability of a vector space with the descriptive precision of discrete categories, providing a robust solution for multimodal music emotion recognition.





Neuroscientific studies support this hybrid model. High-arousal emotions (tension, power, joy) consistently engage sensory and motor areas, whereas low-arousal emotions (peacefulness, nostalgia, sadness) selectively recruit the ventromedial prefrontal cortex (vmPFC) and the hippocampus. Positive-valence emotions engage the reward network (ventral and dorsal striatum, orbitofrontal cortex, and insula), with left-sided lateralization for high-arousal states (joy) and right-sided lateralization for low-arousal, reflective states (nostalgia, tenderness). The right parahippocampal cortex is active in all states except positive, high-arousal conditions, showing that different emotional states rely on distinct neural pathways.

For automated classification, the system processes three primary signals:

- **Lyrics:** Transformer-based models (e.g., BERT, XLNet) map text to semantic vectors to extract valence and identify specific narrative themes. Lyrics are crucial for identifying negative-valence states.
- **Genre and Artist Metadata:** These serve as statistical priors to narrow down the target coordinates.
- **Acoustic Signals:** Digital signal processing (DSP) extracts low-level features such as tempo, mode, roughness, and spectral centroid to calculate valence and arousal coordinates.

Because songs can have contrasting acoustic and lyrical profiles (e.g., upbeat music with dark lyrics), the system must use a late-fusion model. Instead of averaging conflicting values into a neutral score, the system calculates coordinates for each modality and applies a weighting algorithm based on the segment type, allowing the lyrical classifier to dominate during vocal-heavy verses and the acoustic classifier to dominate during choruses or instrumental solos.

Proposed Eight-Category Musical Emotion Taxonomy

The proposed taxonomy consists of eight validated categories that span the musical emotion space. Each category is defined by its coordinates, acoustic properties, and lyrical themes.

Master Coordinate and Feature Mapping

Category Name	Valence (V)	Arousal (A)	Key Acoustic Markers	Dominant Keys	Lyrical Themes
Joyful Activation	+0.8	+0.6	Fast tempo, major mode, staccato, high pitch, consonant.	C Major, Bb Major	Celebration, playfulness, movement.
Triumphant / Power	+0.6	+0.8	Fast tempo, major, sharp attack, brassy timbres.	D Major, F# Major	Victory, resilience, strength.

Category Name	Valence (V)	Arousal (A)	Key Acoustic Markers	Dominant Keys	Lyrical Themes
Peacefulness	+0.8	-0.7	Slow tempo, major, soft dynamics, flat pitch, legato.	G Major, F Major	Quietude, release, nature.
Tenderness	+0.7	-0.3	Slow tempo, major, soft dynamics, warm timbres, vibrato.	A Major, Eb Major	Love, intimacy, devotion.
Nostalgia / Longing	+0.2	-0.4	Slow/moderate tempo, minor-to-major, pitch variability.	C Minor, E Minor	Bittersweet memory, passing time.
Sadness / Melancholy	-0.8	-0.6	Slow tempo, minor, legato, low pitch, minor thirds.	F Minor, C# Minor	Grief, absolute loss, hopelessness.
Tension / Anxiety	-0.6	+0.5	Irregular rhythm, minor/atonal, high spectral variation.	G Minor, D# Minor	Paranoia, entrapment, instability.
Anger / Defiance	-0.7	+0.8	Fast tempo, minor, distortion, vocal screaming.	B Major, Bb Minor	Rebellion, direct hostility, rage.

Technical Profiles of the Taxonomy

Joyful Activation

This category represents a state of vibrant, stimulating happiness that encourages physical movement, playfulness, and immediate behavioral approach. Acoustically, it is characterized by a fast tempo, major mode, consonant harmony, staccato articulation, high average pitch, and rising pitch contours. It maps to keys like C Major and Bb Major, which convey simplicity, naivety, and bright cheer. Lyrically, it is defined by themes of celebration, dancing, sensory pleasure, and optimism. Prototypical genre examples include upbeat Pop, Funk, and Electronic Dance Music, where a steady tempo and bright timbres drive the listening experience.

Triumphant / Power

Triumphant / Power represents a highly active state of strength, energetic vitality, heroic pride, and self-efficacy. Acoustically, it relies on fast or medium-fast tempos, major or mixolydian modes, loud average dynamics, brassy or resonant timbres, sharp envelope attacks, and triumphant, rising pitch contours. It is associated with keys like D Major and F# Major, which convey conquest, victory, and sighs of relief. Lyrically, it features themes of resilience,

overcoming obstacles, power, and defiance of limits. This state is common in Arena Rock, Orchestral Film Scores, and Synth-wave.

Peacefulness / Serenity

This category represents a low-arousal, positive-valence state of cognitive relaxation, tranquility, and meditative release. Acoustically, it is defined by a slow tempo, major mode, consonant harmony, soft dynamics, legato articulation, low pitch, and flat pitch contours. It is associated with keys like G Major and F Major, which convey calmness, gratitude, and controlled composure. Lyrically, it focuses on release, quietude, nature, and breathing. It is common in Ambient, New Age, and Minimalist Classical music, where slow pad textures and a lack of rhythmic percussion create a continuous sonic space.

Tenderness

Tenderness represents a warm, low-activation positive state centering on love, intimacy, interpersonal affection, and protective care. Acoustically, it uses slow-to-moderate tempos, major mode, soft dynamics, gentle attack envelopes, mellow timbres with a high spectral slope, and micro-variations in pitch and timing (vibrato). It aligns with keys like A Major and Eb Major, which convey open-hearted intimacy and honest communion. Lyrically, it uses interpersonal pronouns and themes of devotion, affection, and physical closeness. It is common in Singer-Songwriter tracks, Classic Soul, and R&B Ballads.

Nostalgia / Longing

This category represents a bittersweet, complex emotional state characterized by yearning, reflective memory, and a dual-valence appreciation of absent beauty. Acoustically, it is characterized by slow-to-moderate tempos, minor scales with frequent resolutions to major chords, high pitch variability in solo instruments, soft dynamics, and expressive, sighing intervals. It is represented by keys like C Minor and E Minor, which carry restless yearning and lamentations of lost love. Lyrically, it centers on bittersweet memories, the passing of time, and unresolved desire for an absent target. It is common in Indie Folk, Classic Country, and Jazz Ballads.

Sadness / Melancholy

Sadness / Melancholy is an unambiguous low-valence, low-arousal state of grief, sorrow, and solemn resignation. Acoustically, it is defined by a slow tempo, minor mode, dissonant harmonic intervals, soft dynamics, legato articulation, and a narrow pitch range with descending contours. It heavily utilizes the minor third interval, which mirrors the pitch contours of sad human speech. It maps to keys like F Minor and C# Minor, which convey deep depression, lamentation over loss, and despair. Lyrically, it focus on absolute loss, hopelessness, and grief. It is common in Blues, Doom Metal, and Singer-Songwriter ballads.

Tension / Anxiety

This category represents a state of nervous apprehension, cognitive instability, and psychological unease. Acoustically, it uses irregular or syncopated rhythms, minor or atonal

modes, high spectral variation, gentle attacks with slow build-ups, and sudden dynamic spikes. It maps to keys like G Minor and D# Minor, which carry apprehension, existential angst, and future worry. Lyrically, it focuses on paranoia, confinement, loss of control, and fear. It is common in Horror Scores, Industrial, and Post-Punk, where scraping timbres and unpredictable dynamic changes create a sense of instability.

Anger / Defiance

Anger / Defiance is a high-arousal, low-valence state of intense hostility, systemic rebellion, and aggressive release. Acoustically, it is characterized by fast tempos, minor modes, high acoustic roughness (distortion), harsh articulation, loud dynamics, and irregular spectral flux. It maps to keys like B Major and Bb Minor, which convey rage, fury, and mocking hostility. Lyrically, it focuses on socio-political rebellion, personal rage, and active confrontation. It is common in Heavy Metal, Hardcore Punk, and Aggressive Rap.

Diagnostic Resolution of Emotional Boundary Cases

Distinguishing between adjacent emotional states is a primary challenge in automated music emotion classification. To resolve these ambiguities, specific acoustic and lyrical rules must be established.

Longing vs. Sadness

While both states feature slow tempos and minor modes, they serve different psychological and emotional functions.

- **Lyrical Differentiation:** Sadness lyrics focus on finality, absolute loss, and resignation ("it is over," "nothing left to save," "I am broken"). Longing lyrics are interrogative and future-oriented, focusing on desire and an absent target ("where are you," "will we meet again," "I wait for you"). Contrasts between sad music with versus without lyrics show that lyrics activate the parahippocampal gyrus, amygdala, and Broca's area, highlighting that sad music relies heavily on lyrical content to convey its specific emotional meaning.
- **Sonic Differentiation:** Sadness is defined by a narrow pitch range, low pitch variability, and a strict adherence to minor mode and falling pitch contours. Longing maintains higher pitch variability and utilizes instruments that mimic a weary human voice (such as the pedal steel guitar, muted trumpet, or slide violin). It also frequently features transitions between minor and major chords (such as a minor iv to major I progression) to capture a bittersweet quality.

Power vs. Anger/Defiance

Both states occupy the high-arousal, energetic portion of the affective space but diverge sharply on the valence axis.

- **Lyrical Differentiation:** Power lyrics focus on self-efficacy, resilience, and overcoming obstacles ("we will rise," "stronger," "unbroken"). Anger lyrics focus on destruction, active hostility, and rage ("burn it down," "destroy," "hate").
- **Sonic Differentiation:** Power relies on consonant, stable harmonies, steady rhythms, major or mixolydian modes, bright resonant timbres, and clean vocals. Anger relies on

acoustic roughness (distortion, screaming vocals), harmonic dissonance (minor second, tritone), irregular syncopated rhythms, and high spectral flux.

Tenderness vs. Peacefulness/Serenity

These positive-valence, low-arousal states occupy similar quadrants but differ in their social and acoustic density.

- **Lyrical Differentiation:** Tenderness focus on interpersonal connection, warmth, and relational devotion ("you and I," "hold you close," "love you forever"). Peacefulness focus on solitary, environmental, or meditative states ("silent waves," "rest now").
- **Sonic Differentiation:** Tenderness has a slightly higher arousal profile (≈ -0.3) and features expressive micro-timing variations (vibrato), mellow timbres with slow attack envelopes, and melodic contours that mimic intimate speech. Peacefulness has the absolute lowest arousal (≈ -0.7) and utilizes highly regular, repetitive, or drone-like structural features, devoid of rhythmic percussion or vocal/instrumental vibrato, resulting in a flat, predictable acoustic flow.

Methodological Limitations and Theoretical Challenges

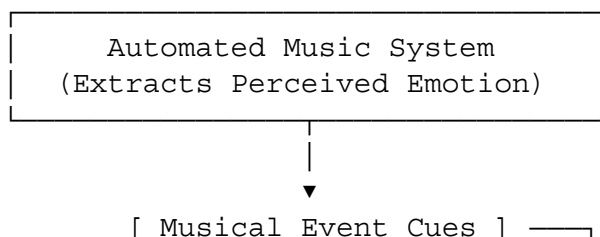
Implementing the proposed framework requires addressing several theoretical limitations and computational challenges in music psychology.

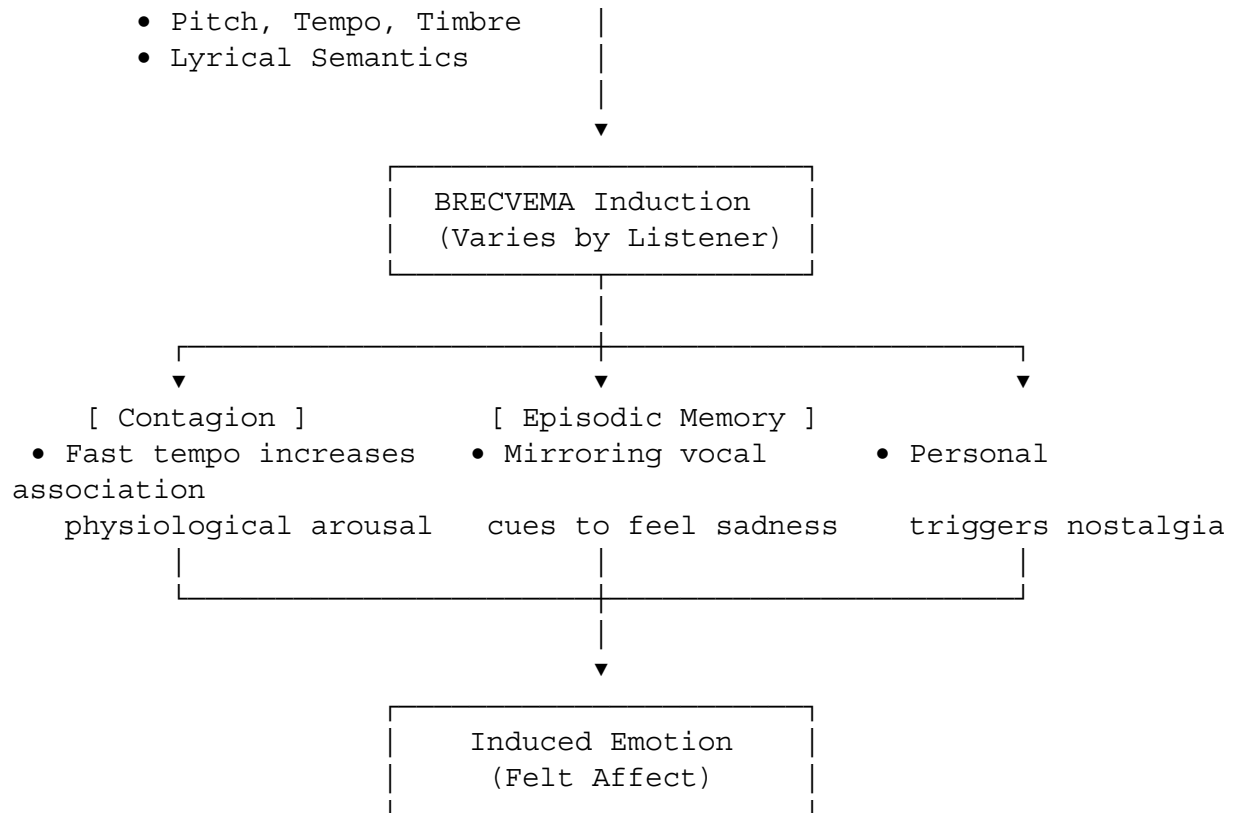
Positive-Valence Bias of Music-Specific Scales

Because music is primarily consumed for pleasure, music-specific taxonomies (such as GEMS and GEMIAC) are heavily biased toward positive, aesthetic emotions. Consequently, basic negative emotions like anger or fear are underrepresented or collapsed into "tension" or "dysphoria". While GEMIAC's inclusion of boredom and gloomy sadness was an attempt to resolve this, automated systems trying to classify aggressive modern music (metal, punk, drill rap) struggle with scales optimized for classical and aesthetic music.

The Perceived vs. Induced Emotion Divergence

Classifying a track based on its acoustic and lyrical properties identifies *perceived* (expressed) emotion. However, the *induced* (felt) emotion in the listener is governed by complex, highly individual psychological mechanisms. Integrating Juslin's BRECVEMA framework highlights this gap, as mechanisms like episodic memory, visual imagery, rhythmic entrainment, or aesthetic judgment cannot be predicted by analyzing a song's audio files or lyrics alone; they depend entirely on the listener's personal history, current cognitive state, and situational context.





An automated system can analyze acoustic markers to detect *perceived* sadness, but the *induced* reaction can be positive, such as when a sad song provides a sense of acceptance and comfort.

The Irony and Multimodal Alignment Problem

Combining lyrics and audio features often results in conflicting data. Upbeat tracks with dark lyrics are common, and a simple averaging of these features in the VA space yields a flat, neutral coordinate, misrepresenting the active tension intended by the artist. Automated models must incorporate dynamic modal weighting, allowing the lyrical classifier to override valence calculations in narrative-heavy genres, while letting acoustic features dictate arousal and valence in high-energy or instrumental passages.

Cultural and Linguistic Relativity

Acoustic codes like minor thirds for sadness or fast tempo for joy are highly validated in Western music and language, but may not generalize globally. A global recommendation tool must adjust its prior coordinates based on region, culture, and localized genre contexts.

Conclusions and Actionable Recommendations

To build a robust software tool for mapping emotional trajectories from musical listening data, the following structural implementation steps are recommended:

- **Adopt a Hybrid Vector Space:** Map continuous Valence-Arousal (VA) trajectories to maintain mathematical tractability for user profile tracking, while populating this space with the proposed eight-category taxonomy for user-friendly labels.
- **Implement Multi-Modal Late Fusion:** Process audio features, lyrics, and metadata separately. Use transformers (e.g., BERT or XLNet) for lyrics to capture negative valence and complex narrative themes, and apply digital signal processing (DSP) to audio for tempo, mode, and spectral roughness to map arousal.
- **Apply Contextual Prior Weights:** Treat genre and artist metadata as statistical priors, adjusting the default classification boundaries based on the regional and cultural context of the music.
- **Differentiate Ambiguous Boundary Cases:** Use explicit, operationalized guidelines to separate tricky boundary cases—such as using interrogative lyrical structures to distinguish longing from sadness, and using acoustic roughness and harmonic dissonance to separate anger from triumphant power.
- **Acknowledge the Perceived-Induced Gap:** Design user-facing interfaces to reflect that the system tracks the *expressed* emotional properties of the music, as the *felt* emotional experience is highly personalized and dependent on the listener's memory and situation.

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